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B.Tech. DEGREE EXAMINATION, APRIL/MAY 2017.

Sixth/Seventh Semester

Information Technology/Computer Science Engineering

ARTIFICIAL INTELLIGENCE

Time : Three hours

Maximum : 75 marks

PART A — ($10 \times 2 = 20$ marks)

Answer ALL questions.

All questions carry equal marks.

1. When is the idea of AI emerged and how?
2. Denote the importance of Means Ends Analysis.
3. Represent the purpose of Propositional Logic.
4. Give a note on Unification.
5. State Bayes rule.
6. Write about Dempster.
7. How to perform the state space search?
8. How would you differentiate a normal and a decision tree?

9. What is the purpose of Minimax search procedure?
10. What is an Expert System?

PART B — (5 × 11 = 55 marks)

Answer ALL questions, ONE from each unit.

All questions carry equal marks.

UNIT I

11. Write in detail about the Heuristic Search Techniques.

Or

12. Depict the structure of agents and highlight its functions.

UNIT II

13. Give a detailed account on Forward and Backward reasoning.

Or

14. Compare and analyze the Weak slot and Strong slot filler structure.

UNIT III

15. Deliver the logics of non-monotonic reasoning in detail.

Or

16. Give a detailed account on Fuzzy Logic.

UNIT IV

17. Differentiate Conditional Planning from Continuous Planning.

Or

18. Elaborate on the significance of Neural Learning.

UNIT V

19. How to add alpha-beta cutoffs in Game programming? Explain.

Or

20. Describe about the Application domains of Robotics.

16. (a) Draw a Bayesian network showing both the topology and the conditional probability tables. (4)
(b) Discuss any two exact inference methods in Bayesian networks. (7)

UNIT IV

17. (a) Explain the terms ordering constraints, causal links, linearization with respect to partial-order planning with an example. (6)
(b) Discuss backward state-space search with an example. (5)

Or

18. (a) Explain neural network – a statistical learning method in detail. (6)
(b) Explain multi agent planning with example. (5)

UNIT V

19. (a) Give the advantages of expert system. Explain it with the help of a case study. (6)
(b) Discuss the different phases of knowledge acquisition. (5)

Or

20. (a) Explain alpha beta pruning with an example. (6)
(b) Explain Minimax search procedure with an example. (5)

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B.Tech. DEGREE EXAMINATION, NOVEMBER 2017.

Seventh Semester

Computer Science and Engineering

ARTIFICIAL INTELLIGENCE

(2013–14 onwards)

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. What are the performance measures in problem solving?
2. State the names of informed search algorithms.
3. State the advantage and disadvantage of semantic net.
4. Differentiate representational adequacy and inferential adequacy.
5. What is non monotonic reasoning?
6. State the significance of fuzzy logic.
7. How Closed-world assumption is useful in representation of states in planning problems?

8. Mention the reason that causes overfitting and technique that can be used to avoid it in decision trees.
9. List the benefits of incorporating artificial intelligence concepts in game playing.
10. State any two real time applications domains for Robotic technology.

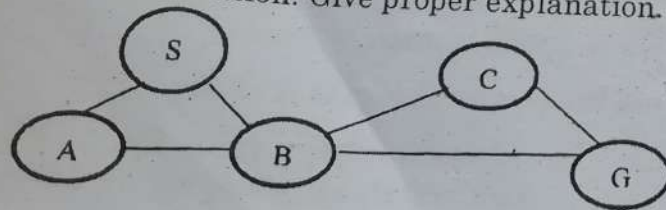
PART B — ($5 \times 11 = 55$ marks)

Answer ALL questions, choosing ONE from each Unit.

All questions carry equal marks.

UNIT I

11. (a) Explain Depth-First search and Depth-Limited search strategies and compare their performances. (6)
(b) Explain briefly backtracking search for Constraint Satisfaction problems. (5)
- Or
12. (a) Find solution using A* algorithm considering the following example. Distance SA-1, SB-1, AB-9, BC-6, CG-5, BG-12. Heuristic values : AG-10, BG-9, CG-5. S is the source, G is destination. Give proper explanation. (6)



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- (b) Discuss about the different Environments with example of each kind of environment.

UNIT II

13. Consider the following sentences :
Whoever can read is literate.
Dolphins are not literate.
Flipper is an intelligent dolphin.

- (a) Translate these sentences into formulas in predicate logic.
- (b) Convert the formulas into clause form.
- (c) Who is intelligent but cannot read - Find using resolution.

Or

14. (a) Discuss the approaches and issues in knowledge representation. (6)
(b) Discuss how knowledge can be represented and inferencing can be done using frames. (3)

UNIT III

15. (a) State Bayes rule and its use. (4)
(b) Discuss inference using full joint distributions. (7)

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B.Tech. DEGREE EXAMINATION,
NOVEMBER/DECEMBER 2016.

Seventh Semester

Computer Science and Engineering/
Information Technology

ARTIFICIAL INTELLIGENCE

(2013–2014 onwards)

Time : Three hours

Maximum : 75 marks

PART A — ($10 \times 2 = 20$ marks)

Answer ALL questions.

All questions carry equal marks.

1. What is Artificial Intelligence?
2. Explain problem reduction.
3. List any two approaches to knowledge representation.
4. Define unification process.
5. State Bayes rule.
6. What is Shafer theory?
7. Define conditional planning.

8. What is inductive learning?
9. What is meant by knowledge acquisition?
10. List out any two application domains of expert systems.

PART B — (5 × 11 = 55 marks)

Answer ALL questions, choosing ONE from each Unit.

All questions carry equal marks.

UNIT I

11. Discuss Heuristic Search technique with an example.

Or

12. Explain the structure of agents and their functions.

UNIT II

13. Explain forward and backward reasoning with appropriate example.

Or

14. Distinguish between strong slot filler structure and weak a lot filler structure with suitable example.

UNIT III

15. Explain certainty factors and rule based systems.

Or

16. Write short notes on fuzzy logic with its appropriate applications.

UNIT IV

17. Discuss multi-agent planning in detail.

Or

18. Discuss Genetic learning with an example.

UNIT V

19. Explain Minimax search procedure in detail.

Or

20. Write short notes on Robotic perception.

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B.Tech. DEGREE EXAMINATION, MAY 2018.

Seventh Semester

Computer Science and Engineering

ARTIFICIAL INTELLIGENCE

(2013-14 onwards)

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL the questions.

1. What is Artificial Intelligence?
2. Define Heuristic search.
3. Define Predicate logic.
4. Differentiate forward Vs backward reasoning.
5. What is meant by Bayesian Networks?
6. What are facts and representations?
7. What do you mean by game Playing?

8. Write a note on Continuous Planning.
9. Discuss the usage of membership functions in fuzzy logic.
10. Write a note about MYCIN Expert system.

PART B — (5 × 11 = 55 marks)

Answer ALL questions.

UNIT I

11. How to define the problem as a state space search? Explain with chess game example.

Or

12. Explain the best first search and breadth first search algorithms with suitable examples.

UNIT II

13. Write a note on the following :
 - (a) Computable functions and predicates. (6)
 - (b) Resolution. (5)

Or

14. What are the different approaches available for knowledge representations? Explain.

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UNIT III

15. Explain about the Bayesian networks.
- Or
16. Write a note on Fuzzy logic with its appropriate applications.

UNIT IV

17. Discuss about the Genetic learning with an examples.

Or

18. Explain about the Conditional Planning.

UNIT V

19. Explain the Architecture of an Expert system.

Or

20. Explain minimax search procedure in details.

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B.Tech. DEGREE EXAMINATION, NOVEMBER 2018.

Seventh Semester

Computer Science and Engineering

ARTIFICIAL INTELLIGENCE

(2013-14 Onwards batches)

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

1. List the fields that form the basis for AI.
2. What is means-end analysis?
3. Mention some of the knowledge representation techniques.
4. What is the role of unification in resolution?
5. Why does uncertainty arise?
6. How will you approach uncertainty through Bayesian networks?
7. What are the different types of planning?

8. What are Recurrent Neural Networks? Give an example.
9. What would happen if robot arm performs an action in blocks world?

10. What is the role of α and β parameters in alpha-beta pruning?

PART B — (5 × 11 = 55 marks)

Answer ALL questions, ONE from each unit.

UNIT I

11. Consider a state space where the start state is number 1 and the successor function for state n returns two states, numbers $2n$ and $2n + 1$. Draw the state space graph. What nodes will be visited for breadth-first search, depth-limited search with limit 3, and iterative deepening search? Would bidirectional search be appropriate for this problem? If so, describe the portion of the state space for states 1 to 15 where appropriate. What is the branching factor in each direction of the bidirectional search?

Or

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12. Describe about PEAS description of agent based tasks. Give a PEAS description of the task environment for the following agents :

- (a) A mobile robot that delivers coffee in an office building
- (b) A crossword puzzle solver
- (c) Spam mail detector

For each of the agents above, characterize the environment according to whether it is fully or partially observable, deterministic or stochastic, episodic or sequential, static or dynamic, discrete or continuous, and single or multi-agent.

UNIT II

13. Give resolution proof for example problem statement :
 - (a) "West is a criminal"
 - (b) Curiosity killed the cat.

Or

14. Explain forward chaining and Backward chaining with an example.

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